

IDS INCIDENT REPORT

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Environment: Kali Linux VM, Snort 3.12.2.0, lab network 192.168.10.0/24

1. Summary

This project involved deploying Snort (an open-source IDS) on a Kali Linux VM, configuring it to monitor a lab network, writing custom detection rules, and validating detection by simulating a network reconnaissance scan. Activity was confirmed at both the alert level (Snort) and the packet level (Wireshark).

Objective:

Deploy Snort.

Detect a simulated port scan.

Analyze the underlying packet traffic to confirm that the detection was accurate.

Result: Successful. Snort generated an alert for the simulated scan and Wireshark packet capture confirmed the traffic pattern matched a TCP SYN scan.

2. Environment Setup

IDS host: Kali Linux VM

IDS software: Snort++ 3.12.2.0

HOME_NET: 192.168.10.0/24

Monitored interface: eth0

Simulated attacker: Windows host, using Nmap

3. Detection Rules

Two custom rules were written and placed in /etc/snort/rules/local.rules, then loaded into Snort's configuration via the ips.include setting in snort.lua.

```
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
```

```
alert tcp any any -> $HOME_NET any (msg:"Possible Nmap TCP Scan"; flags:S; sid:1000002; rev:1;)
```

Rule 1 detects any ICMP (ping) traffic directed at the protected network.

Rule 2 detects TCP packets with only the SYN flag set, the signature behavior of a SYN ("stealth") port scan, which never completes a full TCP handshake.

4. Detection Test

Snort started with a scan run from a separate host.

Snort generated a "Possible Nmap TCP Scan" alert. 1,000 ports scanned, all ports reported closed, 1,019 packets sent in 9.47 seconds.

5. Packet Analysis (Wireshark)

Nmap scan was captured live in Wireshark on eth0. The captured was filtered using "tcp.flags.syn==1 and tcp.flags.ack==0"

This isolated the scan's SYN packets from other traffic.

Observations: one source IP hitting many destination ports rapidly, each packet SYN-only with no completed handshake, which is a packet-level fingerprint of a TCP SYN scan.

6. MITRE ATT&CK Mapping

T1046 (Network Service Scanning): probing a target's ports/services to identify entry points. Both the Snort alert and packet capture independently confirm this behavior.

7. Conclusion

This exercise validated a working, end-to-end detection pipeline:

1. Snort deployed and configured with a defined HOME_NET.
2. Custom detection rules written for ICMP and TCP SYN scan behavior.
3. Simulated attacker traffic generated (Nmap).
4. Traffic correctly detected and alerted on by Snort.
5. Traffic independently confirmed at the packet level via Wireshark.